第 6 組 期末報告企畫書

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一、資料來源

我們使用 Kaggle 網站上的公開數據—「2024 筆記型電腦銷售價格預測」作為期末報告的分析資料(https://www.kaggle.com/datasets/siddiquifaiznaeem/laptop-sales-price-prediction-dataset-2024)。

二、資料簡述

此資料共有 1020 筆，每一筆數據代表一種筆電的機型。我們的目標是使用特徵（例如：品牌、記憶體容量、儲存空間大小、處理器、顯卡、螢幕大小等等）建立模型，去預測筆電的銷售價格。

各變數的名稱、變數類型、類別數量（only 類別型變數）、遺失率如下表（各變數的唯一值放在「三、預計分析方法」後面）：

No.	Variable	Variable Type C: Continuous N: Nominal O: Ordinal B: Binary	# of category (excluding NA)	Missing Rate (%)
1	X	—	—	—
2	Name	—	—	—
3	Brand	N	31	0
4	Price	C	—	0
5	Rating	C	—	0
6	Processor_brand	N	7	0
7	Processor_name	N	36	0
8	Processor_variant	N	125	2.35
9	Processor_gen	O	13	12.64
10	Core_per_processor	O	12	1.18
11	Total_processor	O	8	43.82
12	Execution_units	O	5	43.82
13	Low_Power_cores	B	2	0
14	Energy_Efficient_Units	B	2	0
15	Threads	O	14	4.71
16	RAM_GB	O	10	0
17	RAM_type	N	12	2.16
18	Storage_capacity_GB	O	10	0

19	Storage_type	N	4	0
20	Graphics_name	N	136	0.20
21	Graphics_brand	N	7	0.20
22	Graphics_GB	O	6	63.92
23	Graphics_integrated	B	2	0.20
24	Display_size_inches	O	22	0
25	Horizontal_pixel	O	22	0
26	Vertical_pixel	O	22	0
27	ppi	C	—	0
28	Touch_screen	B	2	0
29	Operating_system	N	13	0

三、預計分析方法

由於反應變數是筆電價格（Price），屬於連續型變數，因此我們會使用的分析模型包含：(1) Linear Regression、(2) SVM Regressor、(3) Random Forest Regressor (4) Neural Network、(5) XGBoost、(6) Catboost、(7) LightGBM。

遺失值處理，會比較下列方法：(1) Complete case (2) MICE。

評估模型的好壞方法會使用：(1) MAE、(2) RMSE。

```
> unique(laptop$Processor_gen)
 [1]  5  7 12 11 14 NA 13  6  3  8 10  1  9  4
> unique(laptop$Core_per_processor)
 [1]  6  4 12  2 10 24  8 16 14 20 NA 11  5
> unique(laptop$Total_processor)
[1] NA  2  4  8  6 12  5  1 10
> unique(laptop$Execution_units)
[1] NA  4  8 16 12  6
> unique(laptop$Low_Power_Cores)
[1] 0 2
> unique(laptop$Energy_Efficient_Units)
[1] 0 1
> unique(laptop$Threads)
 [1] 12  8 16  4 32 NA 22 24 20 18 14 28  2  6  5
> unique(laptop$RAM_GB)
 [1]  8 16 32  4 48 18 64  2 12 36
> unique(laptop$RAM_type)
 [1] "DDR4"    "LPDDR5"  "LPDDR4"  "DDR5"    NA        "LPDDR4X" "LPDDR5X" "Unified"
 [9] "DDR3"    "LPDDRX4" "DDR6"    "LPDDR3"  "PDDR5X"
> unique(laptop$Storage_type)
[1] "SSD"               "Hard Disk & SSD" "NVMe SSD"        "Hard Disk"
> unique(laptop$Storage_capacity_GB)
 [1]  512 1000  256 1256 2000  128   32   64 4000 1512

> unique(laptop$Brand)
 [1] "HP"         "Lenovo"     "Samsung"    "Tecno"      "Xiaomi"     "Dell"       "Asus"
 [8] "Wings"      "Apple"      "Zebronics"  "Gigabyte"   "Acer"       "MSI"        "Infinix"
[15] "Honor"      "Colorful"   "iBall"      "Walker"     "Chuwi"      "ASUS"       "Ultimus"
[22] "Jio"        "Microsoft"  "Primebook"  "Ninkear"    "Avita"      "Huawei"     "Fujitsu"
[29] "LG"         "Razer"      "AXL"
> unique(laptop$Rating)
 [1] 4.30 4.45 4.65 4.75 4.25 4.35 4.00 4.55 4.70 4.60 4.10 4.20 4.15 4.40 4.05 4.50 3.
> unique(laptop$Processor_brand)
[1] "AMD"       "Intel"     "Apple"     "MediaTek"  "Qualcomm"  "Microsoft" "HiSilicon"
> unique(laptop$Processor_name)
 [1] "AMD Ryzen 5"                 "AMD Ryzen 3"
 [3] "Intel Core i3"               "Intel Core i5"
 [5] "Intel Core i9"               "AMD Ryzen 7"
 [7] "Apple M1"                    "Intel Core i7"
 [9] "Apple M3"                    "Apple M2"
[11] "Intel Core 5"                "Intel Core Ultra"
[13] "AMD Ryzen 9"                 "AMD Athlon"
[15] "Intel Celeron"               "Apple M3 Max"
[17] "Apple M3 Pro"                "Apple M2 Apple M2 Chip"
[19] "eration Intel Core"          "Intel Atom Quad"
[21] "Intel Core 7"                "MediaTek Kompanio"
[23] "AMD Athlon Silver"           "Qualcomm X Elite"
[25] "Intel Core 3"                "Intel Pentium Gold"
[27] "AMD Athlon Pro"              "Intel"
[29] "Mediatek"                    "MediaTek"
[31] "Intel Core I3-1115G4"        "Microsoft SQ1 SQ1"
[33] "Intel Pentium"               "HiSilicon  Kirin 9006C 9006C"
[35] "Intel Celeron Dual"          "Intel Pentium Silver"
> unique(laptop$Processor_variant)
  [1] "5600H"     "7320U"     "1215U"     "1240P"     "1115G4"    "1235U"     "14900HX"
  [8] "12500H"    "1135G7"    "5300U"     "7840HS"    NA          "11800H"    "12450H"
 [15] "5500U"     "11400H"    "7330U"     "12650H"    "120U"      "155H"      "1220P"
 [22] "7730U"     "13700HX"   "13450HX"   "6900HS"    "12700H"    "125H"      "1155G7"
 [29] "5800H"     "125U"      "14700HX"   "13650HX"   "13900H"    "3050U"     "7530U"
 [36] "8840HS"    "13420H"    "11260H"    "N4020"     "7535HS"    "3250U"     "13500H"
 [43] "7735HS"    "155U"      "6800H"     "1165G7"    "1355U"     "1035G1"    "1255U"
 [50] "N5100"     "1335U"     "L3…"       "10750H"    "Z3735F"    "8845HS"    "150U"
 [57] "7840U"     "8640HS"    "7520U"     "185H"      "7120U"     "1125G4"    "1340P"
 [64] "12900HX"   "1360P"     "Elite"     "100U"      "7505"      "1334U"     "1315U"
 [71] "N4500"     "7735U"     "i7"        "1340p"     "7940HS"    "13500HX"   "3045B"
 [78] "5625U"     "5425U"     "10210U"    "5800U"     "5700U"     "12450HX"   "N100"
 [85] "13900HX"   "1305U"     "13700H"    "MT8788"    "N305"      "5500H"     "5800HS"
 [92] "13620H"    "6600H"     "5600HS"    "13980HX"   "i5"        "11300H"    "1230U"
 [99] "processor" "1195G7"    "10510U"    "SQ1"       "3500U"     "i9"        "8265U"
[106] "8550U"     "10210Y"    "L16G7"     "9006C"     "11320H"    "7840H"     "7745HX"
[113] "6800HS"    "13950HX"   "1005G1"    "1250U"     "1035G4"    "1365U"     "13900HK"
[120] "5825U"     "7845HX"    "N6000"     "N4120"     "12900H"    "1260P"     "5500u"
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```
> unique(laptop$Graphics_brand)
[1] "AMD"     "Intel"   "NVIDIA" "Apple"  "ARM"     "Adreno" "Radeon" NA
> unique(laptop$Graphics_GB)
[1]  4 NA  8  6 16 12  2
> unique(laptop$Graphics_integreted)
[1] FALSE  TRUE    NA
> unique(laptop$Display_size_inches)
 [1] 15.60 13.30 16.10 13.60 15.00 16.00 14.00 13.40 15.30 18.00 16.20 14.20 17.30 11.60
[15] 14.10 17.00 13.50 12.40 14.50 13.00 15.75 12.00
> unique(laptop$Touch_screen)
[1]  TRUE FALSE
> unique(laptop$ppi)
 [1] 141.21 165.63 187.51 136.83 226.98 224.51 146.86 141.51 157.35 212.26 168.97 161.73
[13] 188.68 100.45 211.82 224.22 188.28 251.57 254.02 111.94 235.85 182.44 242.59 183.58
[25] 253.93 254.67 145.14 135.09 127.34 226.42 156.23 169.78 177.58 137.68 255.36 188.73
[37] 129.58 200.84 215.63 267.08 217.71 111.14 148.87 283.02 266.37 225.29 234.22 266.26
[49] 139.87 216.33 185.43 192.48 170.24 213.29 141.31 208.20 188.24 170.93 261.25 229.47
[61] 203.97 337.93
> unique(laptop$Operating_system)
 [1] "Windows 11 OS"     "Mac OS"            "DOS OS"            "Ubuntu OS"
 [5] "Windows 10 OS"     "Chrome OS"         "jio"               "Prime OS"
 [9] "DOS 3.0 OS"        "Android 11 OS"     "Mac 10.15.3\t OS" "Mac Catalina OS"
[13] "Linux OS"
> unique(laptop$Horizontal_pixel)
 [1] 1920 1080 2560 2880 2240 1366 3840 3456 3200 1440 3024 3072 2561 2256 3000 1536 2160
[18] 2048 2520 1280 1200 1600
> unique(laptop$Vertical_pixel)
 [1] 1080 1920 1600 1664 1200 1800 1400  768 1620 1864 1440 2400 2234 2000 2560 1964 2160
[18] 1504 1024 1536 1680 1280
```